

Enterprise Governance & Knowledge Management System (EGKMS)

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Functional Requirements Specification (FRS)

Version: 1.0

Project Type: Enterprise Governance & Knowledge Management Platform

Technology Stack

- Frontend: React.js
 - Backend: Node.js + Express.js
 - Database: MongoDB
 - Authentication: JWT + Refresh Token
 - Storage: AWS S3 (Private Bucket)
 - UI: Material UI
-

1. Project Objective

Purpose

Enterprise Governance & Knowledge Management System (EGKMS) is a centralized platform for managing an organization's strategic knowledge, governance, policies, planning, operational processes, system improvements, SOPs, standards, and continuous improvement initiatives.

The platform is **NOT** intended for:

- Client Proposals
- Sales Quotations
- Sales Documents
- Project Management Documents
- Software Development Documents
- Technical Project Documentation
- Financial Documents

The platform exists only for organization-level knowledge, governance, planning and continuous organizational improvement.

2. Vision

Create a centralized organizational knowledge repository where every strategic decision, organizational process, policy, SOP, improvement, innovation, and governance document is securely maintained with complete lifecycle management, version control, approval workflow, and auditability.

3. Future Scalability

The architecture must be **Domain Driven**.

Current Domain

- Governance

Future Domains (without changing existing code)

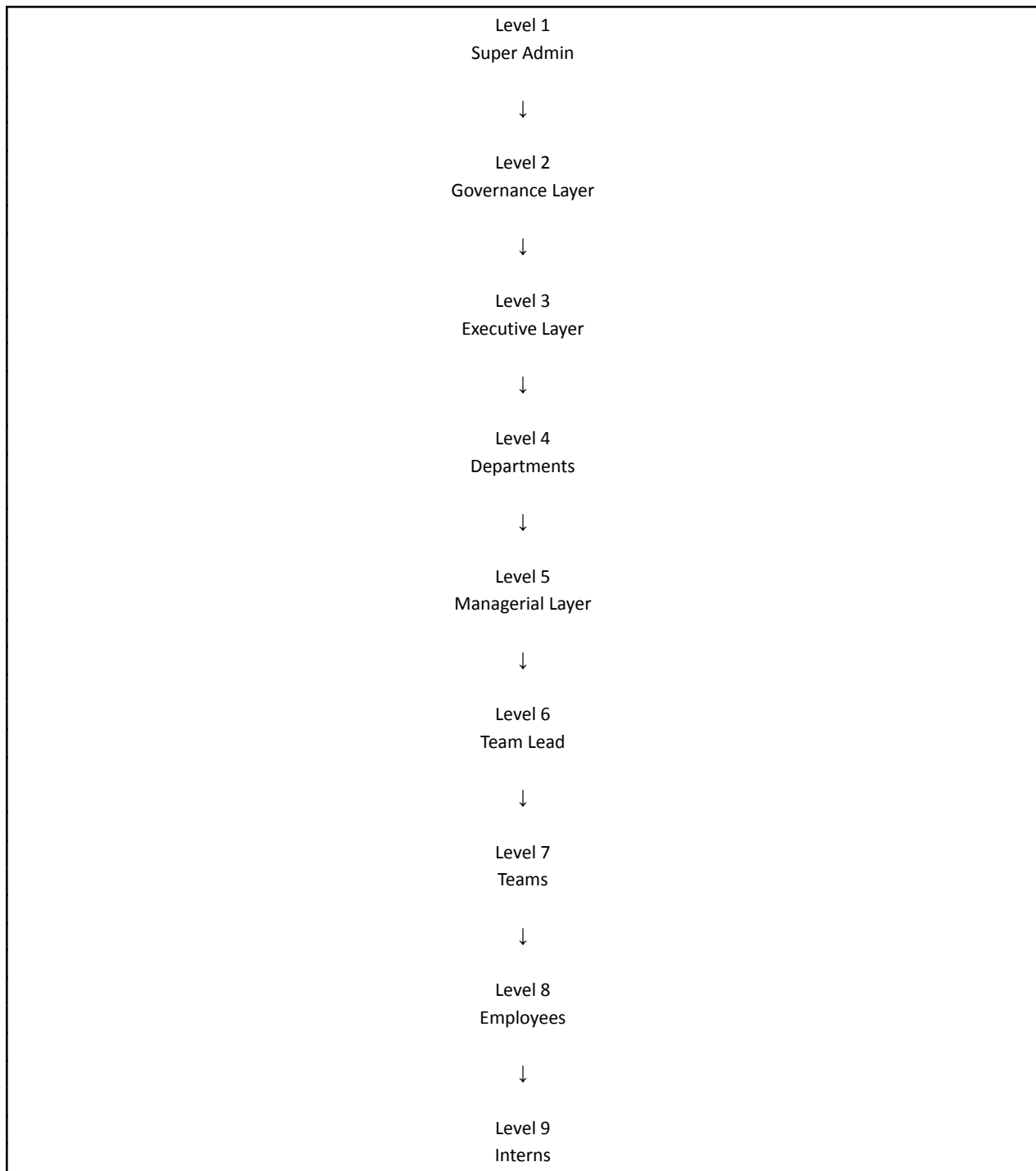
- Project Documents
- Sales Documents
- Finance Documents
- Client Documents
- Legal Documents

Future expansion should only require:

- Creating a new Document Type
- Creating Categories
- Configuring Workflow
- Assigning Permissions

The existing Document Engine, Approval Engine, Version Engine, Permission Engine, Search Engine, and Audit Engine must not require modification.

4. Organizational Hierarchy



Every user belongs to:

- One Hierarchy Level
- One Department

- One Team (Optional)
- One Reporting Manager

5. Role Responsibilities

5.1 Super Admin

Purpose

Platform Owner

Complete control over the entire platform.

Responsibilities

Organization Management

- Create Governance Members
- Create Executive Members
- Create Departments
- Create Department Heads
- Create Managers
- Create Team Leads
- Create Teams
- Create Employees
- Create Interns

User Management

- Create Users
- Update Users
- Disable Users
- Remove Users
- Reset Password
- Transfer Users
- Assign Reporting Manager

Permission Management

Can configure permissions for every hierarchy.

Examples

- View
- Create
- Edit
- Delete
- Review
- Approve
- Publish
- Archive
- Restore

Document Management

Can

- View every document
- Edit every document
- Archive any document
- Permanently delete approved/commenced documents
- Restore archived documents
- Change document ownership
- Change workflow
- Force publish
- Force archive

Monitoring

Super Admin can monitor everything.

Examples

- Login History
- User Activity
- Document Activity
- Approval Activity
- Search History
- Downloads
- Failed Login
- Version Changes
- Comments
- Pending Documents
- Escalations

The Super Admin Dashboard should provide a complete audit of all user activities across the platform.

5.2 Governance Layer

Purpose

Owns organization-wide governance and strategic documentation.

Can Create

- Governance Documents
- Policies
- Strategic Planning
- Standards
- Organization-wide SOPs
- Governance Frameworks

Can Manage

- Executive Layer
- Executive Access
- Executive Permissions (as assigned by Super Admin)

Cannot

- Delete documents created or approved by Super Admin.
- Delete documents published by higher authorities.

Can Delete

Only their own documents while the document status is:

- Draft
- Pending Review
- Returned

Once a document reaches **Approved**, **Published**, or **Commenced**, it becomes immutable for Governance users. Only the Super Admin may delete it.

5.3 Executive Layer

Purpose

Manage organization-wide execution.

Can Create

- Departments
- Department Heads
- Department Documents
- Operational Documents
- Process Improvement Documents

Permissions

Permissions are assigned by Governance or Super Admin.

Cannot

Delete documents published by:

- Governance
- Super Admin

Can Delete

Only their own documents until the document reaches:

- Approved
- Published
- Commenced

After that, only Super Admin can delete.

5.4 Departments / Department Head

Purpose

Manage department knowledge.

Can

- Create Managers
- Configure department repository
- Review department submissions
- Approve department documents

- Submit documents to Executive Layer

Cannot Delete

Documents published by

- Governance
- Super Admin
- Executive Layer

Can Delete

Own documents only until:

- Draft
- Returned
- Pending Approval

After publication or commencement, only Super Admin can delete.

5.5 Managerial Layer

Purpose

Manage operational teams.

Can

- Create Team Leads
- Review Team Documents
- Approve Team Documents
- Return Documents
- Submit to Department Head

Permissions

Assigned by Department Head.

Cannot Delete

Documents approved or published by:

- Governance

- Super Admin
- Executive Layer
- Department Head

Can Delete

Own documents only before approval or commencement.

5.6 Team Lead

Purpose

Manage Teams.

Can

- Create Teams
- Review Employee Documents
- Review Intern Documents
- Submit Documents to Manager
- Assign Team Tasks

Permissions

Assigned by Manager.

Cannot Delete

Documents approved by

- Governance
- Super Admin
- Executive
- Department
- Manager

Can Delete

Own draft documents only before approval.

5.7 Teams

Purpose

Working group inside department.

Can

- Collaborate
- Prepare Documents
- Submit Documents to Team Lead

Permissions

Assigned by Team Lead.

Cannot Delete

Documents approved by higher hierarchy.

Can Delete

Own documents before approval.

5.8 Employee

Permissions

Assigned by:

- Team Lead
- Reporting Manager

Can

- Create Documents
- Edit Own Documents
- Submit Documents
- View Assigned Documents

Cannot approve.

5.9 Intern

Permissions

Assigned by

- Senior Employee
- Team Lead
- Reporting Manager

Can

- Draft Documents
- Submit Drafts
- Edit Own Drafts

Cannot approve.

6. Document Ownership Rule

Every document has exactly one Owner.

Only the Owner can edit the document until it is submitted for review.

After submission:

- Editing follows the approval workflow.
- Version history is maintained.
- Ownership remains unchanged unless transferred by Super Admin.

7. Document Deletion Policy

This rule is mandatory across the platform.

7.1 Draft Documents

Document Creator

Can Delete

Yes

7.2 Under Review

Document Creator

Can Delete

No

Reviewer must return it first.

7.3 Approved

Document Creator

Cannot Delete

Only Super Admin

7.4 Published

Only Super Admin

7.5 Active / Commenced

Only Super Admin

7.6 Archived

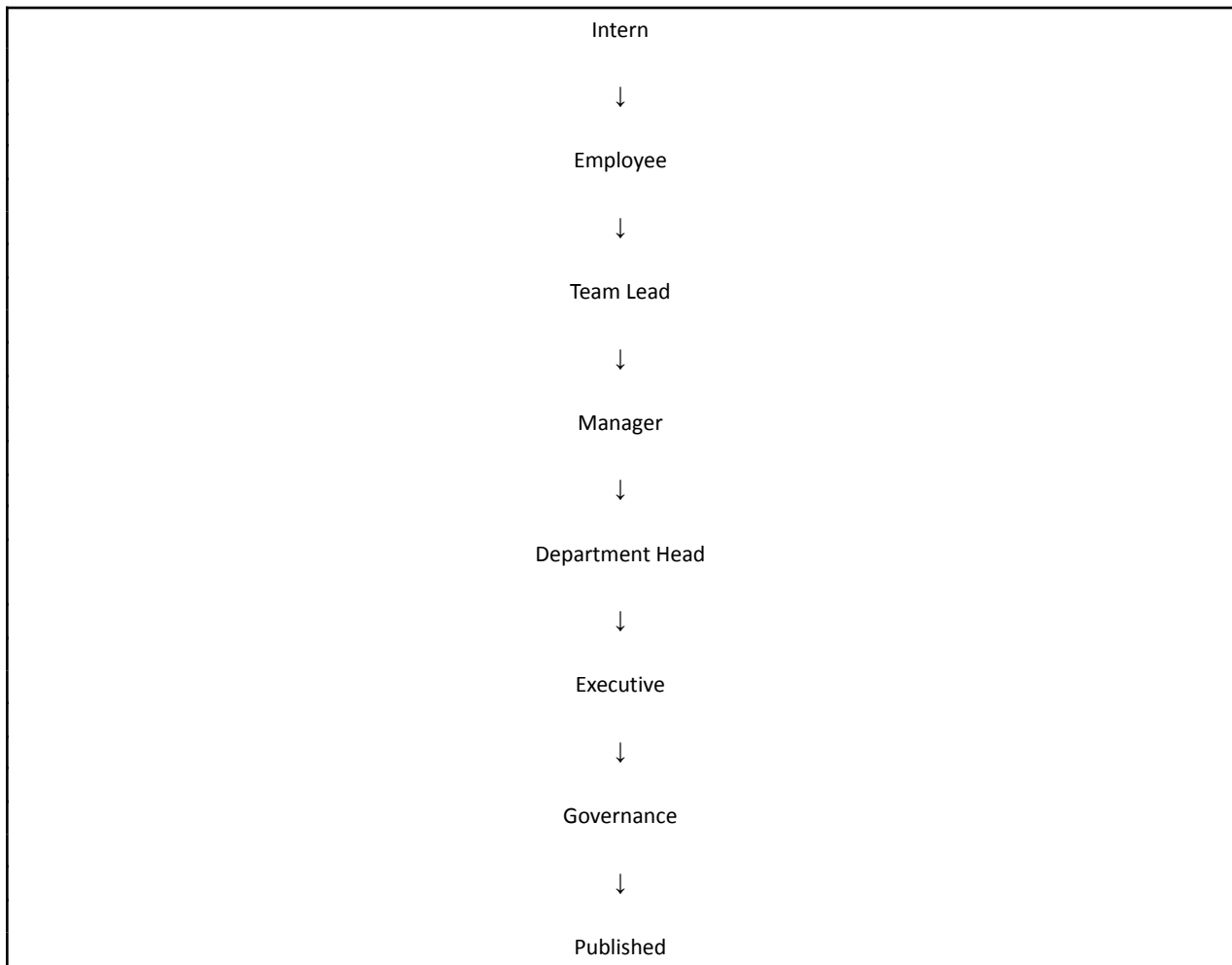
Only Super Admin

No hierarchy can delete documents published by a higher hierarchy.

8. Document Approval Flow

Documents must always follow the organizational hierarchy.

Example

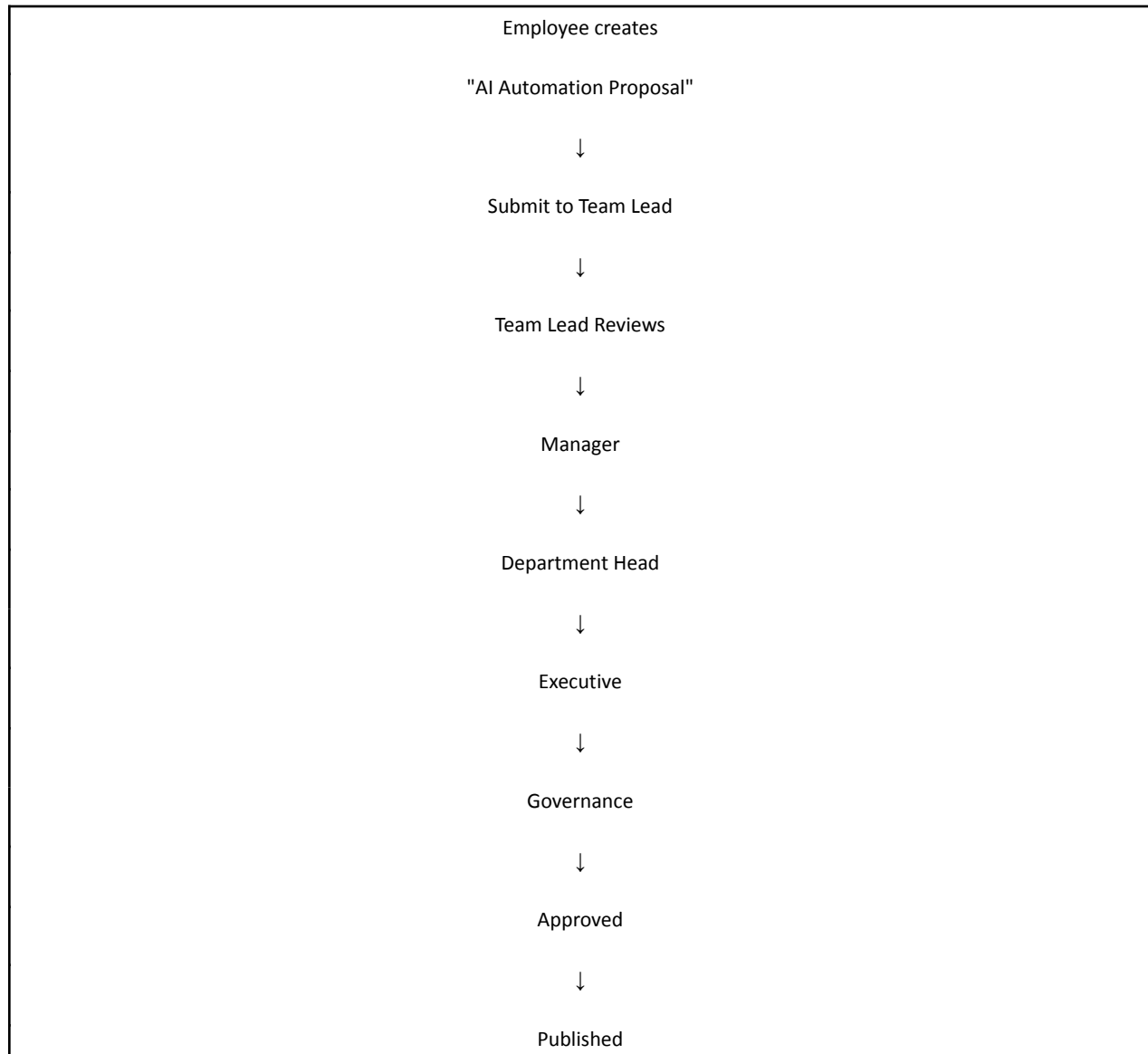


Hierarchy skipping is not allowed.

9. Strategic Idea Submission Flow

This platform is designed to encourage innovation from every organizational level while preserving governance.

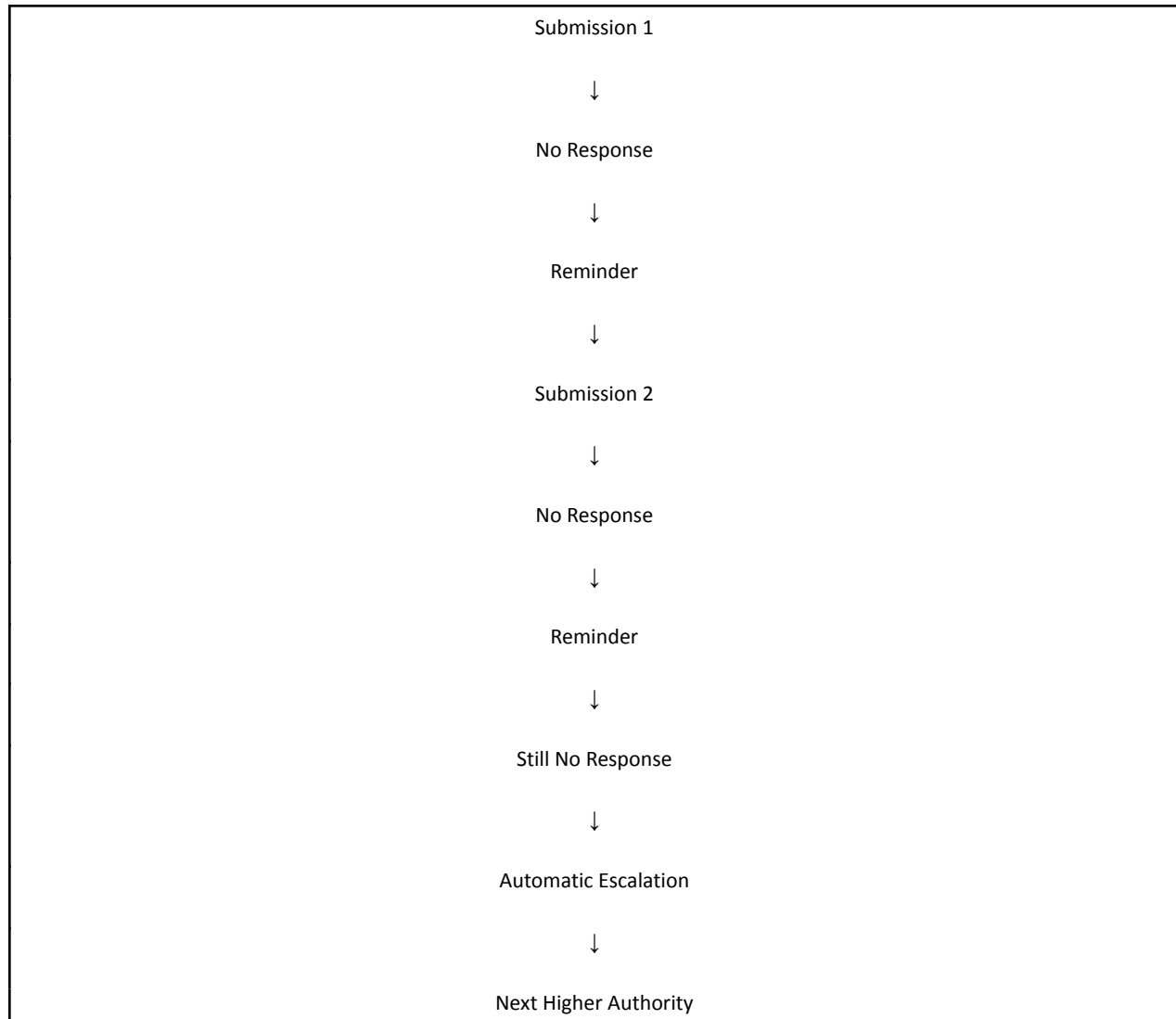
Example



Every strategic idea follows this chain.

10. Escalation Rule

If a document remains unattended:

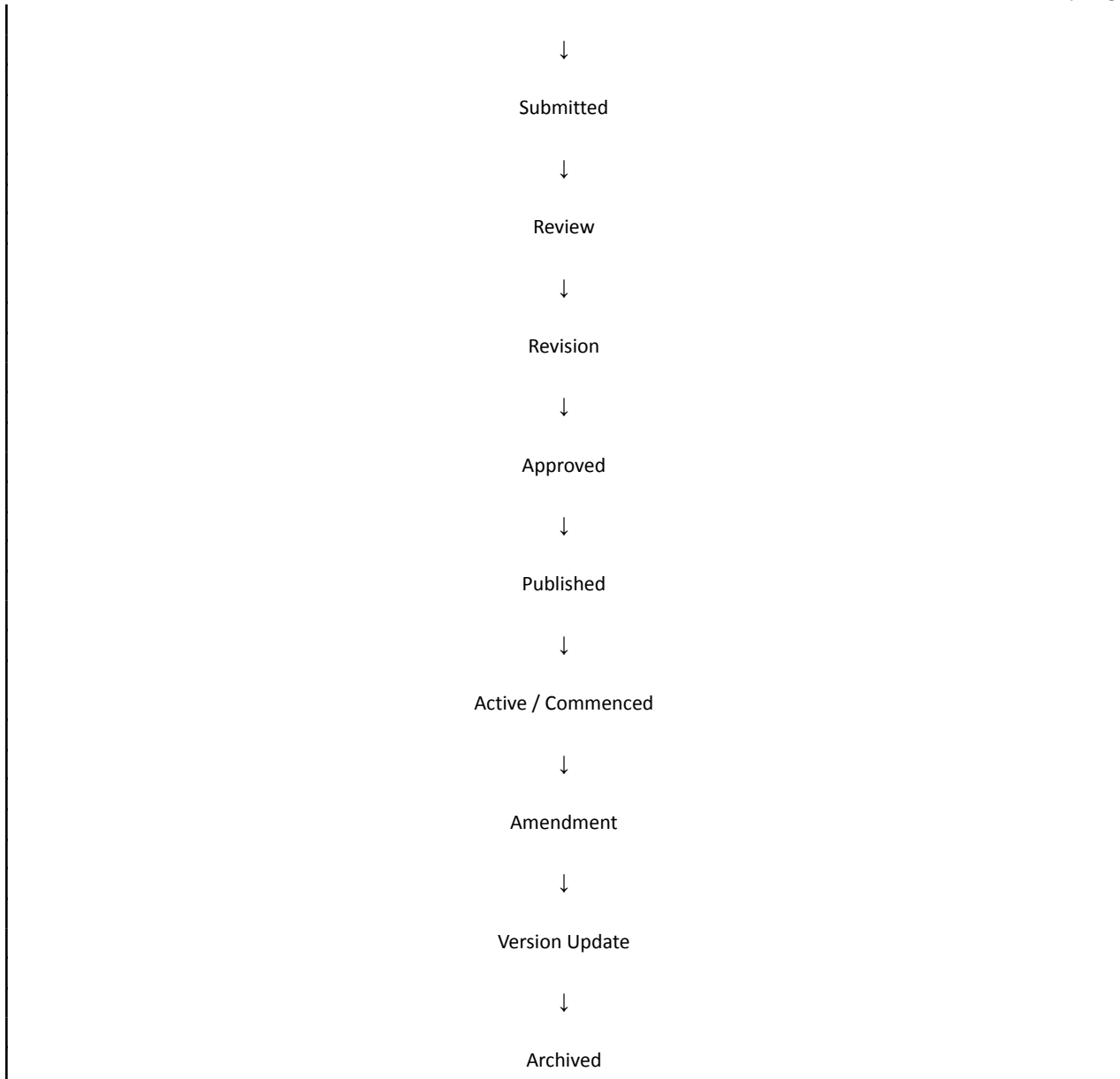


If the document reaches Executive Level and remains unresolved after two reminders, it may be escalated directly to the Super Admin.

This prevents important organizational improvements from being blocked indefinitely while preserving the normal chain of command.

11. Document Lifecycle

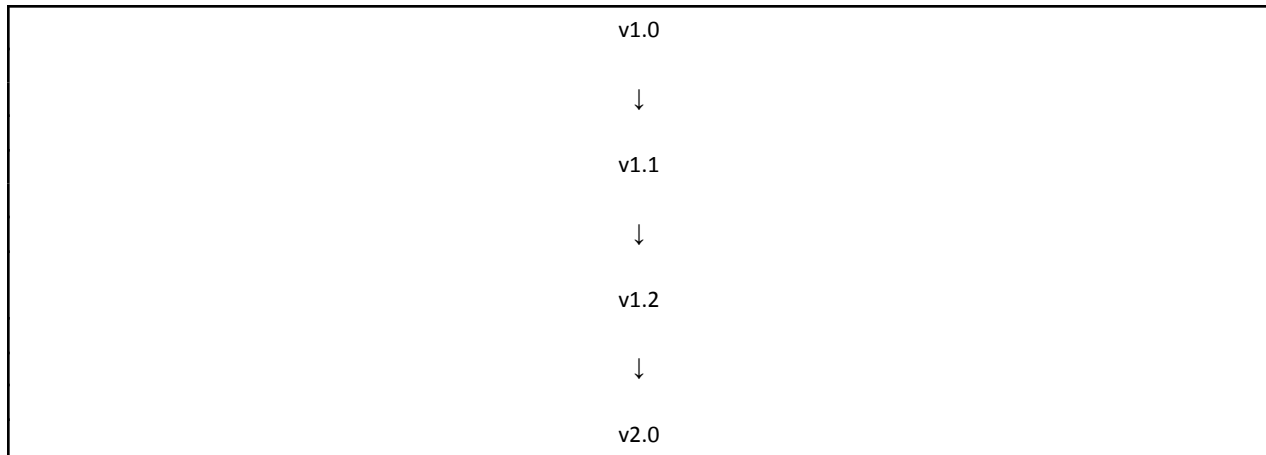
Draft



Every transition must generate an audit log.

12. Version Control

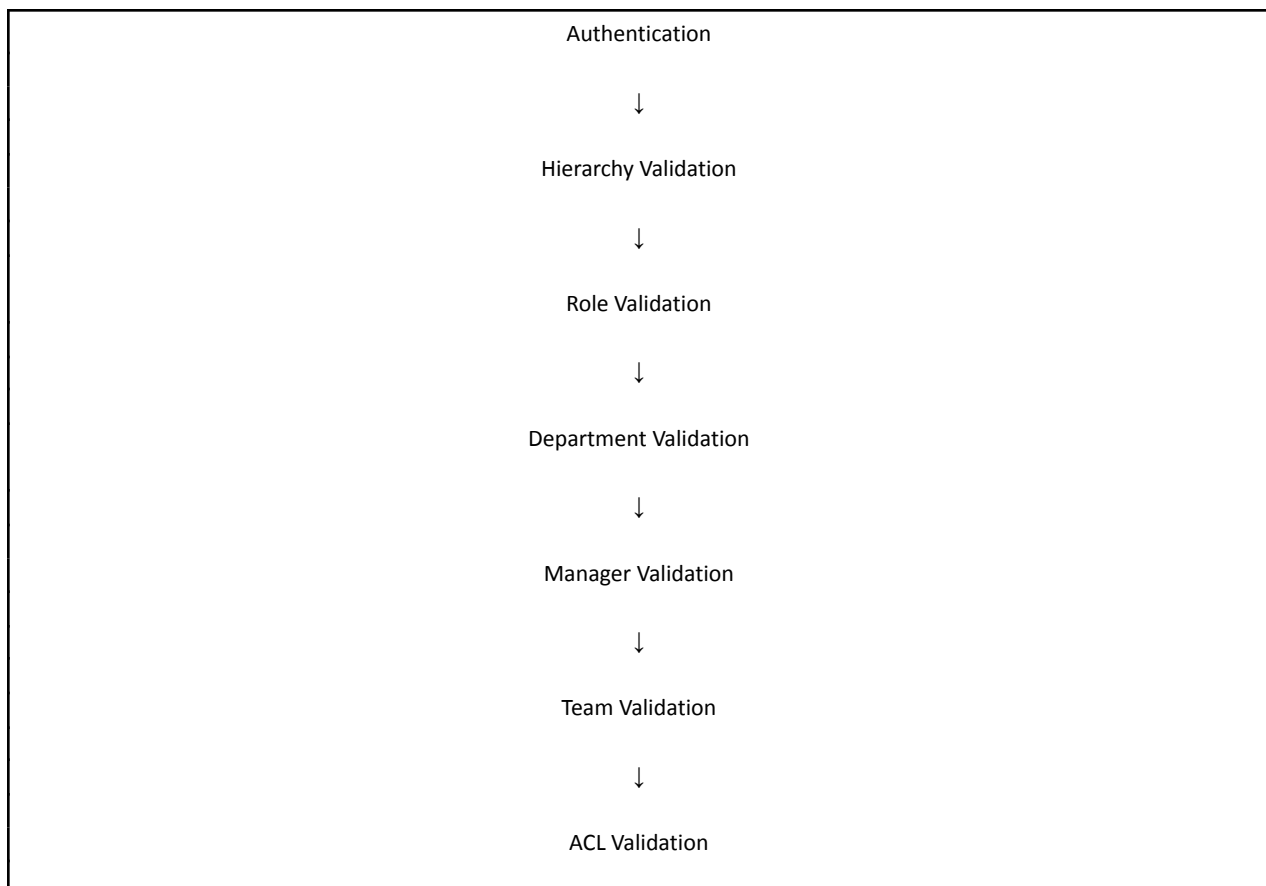
Every modification creates a new version.

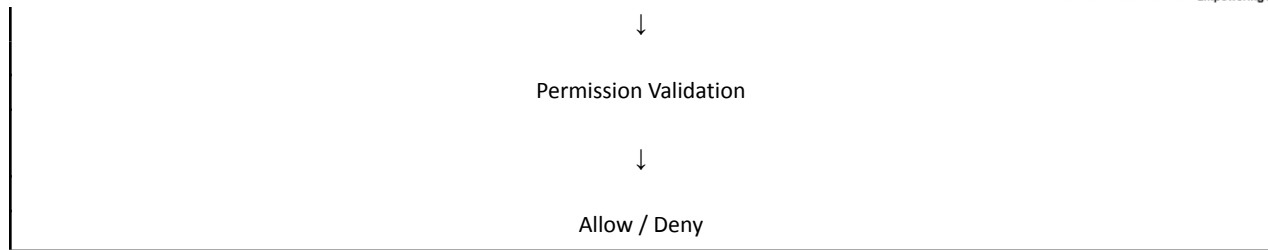


No version is overwritten.

13. Access Control Engine

Every request passes through this sequence:





Access is granted only if every validation succeeds.

14. Audit & Monitoring

Every action is recorded.

Examples:

- Login
- Logout
- Document Created
- Document Viewed
- Document Edited
- Version Created
- Approval
- Rejection
- Comment
- Assignment
- Escalation
- Permission Changes
- User Creation
- Password Reset
- Workflow Changes

Audit logs are immutable and visible only to authorized roles.

15. Analytics Dashboard

Dashboards should be filtered by hierarchy.

Super Admin

- Organization Overview
- All Users

- All Documents
- Pending Approvals
- Workflow Bottlenecks
- Escalations
- Activity Heatmap
- Security Events
- Storage Usage
- Department Statistics

Governance

- Strategic Documents
- Pending Governance Approvals
- Published Governance Documents
- Organization-wide Policy Status

Executive

- Department Performance
- Pending Executive Reviews
- Process Improvement Pipeline

Department Head

- Department Repository
- Team Performance
- Pending Department Reviews

Manager

- Team Review Queue
- Team Activity
- Employee Contributions

Team Lead

- Team Documents
- Employee Drafts
- Intern Submissions

Employee / Intern

- My Drafts
- Assigned Documents
- Submitted Ideas
- Review Status

16. Core Design Principle

The platform is a Governance & Knowledge Platform, not a file storage system. Every document exists to improve the organization's strategy, governance, policies, processes, systems, operational excellence, and long-term growth. Every document must follow the organizational hierarchy, every action must be auditable, and every permission must be enforced through layered access control. The architecture must remain modular so future business domains (Projects, Sales, Finance, Legal, etc.) can be added by configuration without changing the core document, workflow, permission, versioning, or audit engines.